FastVGGT: Training-Free Acceleration of Visual Geometry Transformer

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Abstract

Scaling feed-forward 3D vision models such as VGGT to long image sequences is hindered by quadratic-time global attention and high memory usage, which also exacerbate error accumulation over hundreds to thousands of frames. We diagnose strong redundancy in VGGT's global attention ("token collapse") and introduce FastVGGT, a training-free, 3D-aware token merging scheme that reduces computation while preserving geometric anchors and dense outputs. Our method partitions tokens into (i) reference tokens from the first frame that are never merged, (ii) salient tokens per frame preserved at a fixed stride to maintain correspondences, and (iii) remaining tokens subsampled by region-based random selection for uniform coverage. Within Global Attention, similar source tokens are merged into their nearest destination tokens by cosine-similarity averaging; a deterministic unmerging restores full token resolution before dense prediction. A VRAM-efficient VGGT variant further discards unused intermediates, enabling thousand-frame inputs. Experiments on ScanNet-50, 7 Scenes, and NRGBD show up to 4x speedups with equal or better point cloud and pose accuracy; for example, on ScanNet-50 (1000 images) Chamfer Distance improves from 0.471 to 0.425 while wall-clock time drops from 724.6s to 180.7s.*

1. Introduction

Feed-forward geometry transformers have recently demonstrated impressive capability to estimate camera parameters, depth, and point clouds directly from collections of images without iterative optimization [10]. However, scaling these models to long sequences (hundreds to thousands of frames) is challenged by the quadratic complexity of global self-attention and large VRAM requirements. In practice, naive long-sequence inference often leads to out-of-memory failures and increased drift as small attention errors compound across frames.

We observe that VGGT's global attention maps exhibit striking similarities across tokens at multiple depths, suggesting substantial redundancy. This phenomenon, which we call token collapse, implies that many tokens play near-identical roles in global aggregation. If we can safely merge redundant tokens while preserving a small set of geometrically important anchors, we may cut compute and memory without retraining and without harming geometric accuracy.

We propose FastVGGT, a training-free acceleration of VGGT that introduces a 3D-aware token partitioning and merging mechanism inside Global Attention. Tokens are grouped into non-mergeable anchors from the first frame, salient tokens preserved at a fixed spatial stride in every frame to stabilize correspondences, and residual tokens that are merged using cosine-similarity to nearby anchors selected via region-based sampling. Crucially, we unmerge before the dense predictors to restore full resolution, ensuring that downstream heads operate unchanged.

Our contributions are threefold: (1) we diagnose and visualize redundancy in VGGT's global attention; (2) we introduce the first training-free token merging tailored to feed-forward 3D geometry transformers; and (3) we demonstrate large speedups with maintained or improved accuracy across point cloud reconstruction and pose estimation benchmarks. Figure 1 illustrates the problem setting and overall gains.

Why it works. Global attention aggregates information across all frames. By preserving a small set of geometric anchors (first-frame tokens and fixed-stride salient tokens) and merging only locally redundant tokens into these anchors, FastVGGT maintains cross-view constraints while reducing token count during attention. The deterministic unmerging restores the original token layout for dense prediction, so heads and losses remain compatible.

2. Related Work

Feed-forward 3D geometry transformers. VGGT [10] frames multi-view 3D estimation as a single forward pass that predicts camera parameters, depths, and dense point clouds. Contemporary feed-forward multi-view methods emphasize long-context reasoning and efficiency, including Fast3R [14] and CUT3R [11]. Streaming and long-context extensions (e.g., StreamVGGT) tackle incremental processing [16]. Our work is complementary: we accelerate VGGT inference without retraining and without architectural changes to dense heads.

Token reduction in transformers. Token merging and

Figure 1. FastVGGT processes up to 1,000 images in a single pass, achieving up to 4x faster inference while mitigating long-sequence error accumulation. The figure highlights anchors (first frame), fixed-stride salient tokens, and merged regions that preserve cross-view constraints.

pruning have been widely explored to speed up vision transformers [2]. Diffusion and video transformers also benefit from training-free token merging variants [1]. Beyond these, adaptive or learned reduction has been studied: TokenLearner for videos [7], PatchMerger [6], and clustering-based tokenization [4, 15]. These typically require training or architectural changes; FastVGGT is training-free and geometry-aware.

Multi-view 3D reconstruction from images. Pairwise methods such as DUSt3R [12] and multi-view generalizations (e.g., Fast3R) [14] have advanced feed-forward 3D perception. FastVGGT targets the VGGT family [10], focusing on long-sequence scalability.

3. Method

Our goal is a training-free mechanism that reduces compute in VGGT's Global Attention while preserving geometric fidelity. We first partition tokens, then merge within the attention block, and finally unmerge to recover full resolution before dense prediction.

3.1. 3D-Aware Token Partitioning

Given a sequence of frames tokenized into per-image grids plus special tokens, we partition tokens as follows:

- *First-frame anchors*: all tokens (including specials) from the first frame are never merged to provide a fixed world reference.
- *Salient tokens per frame*: we preserve roughly 10% of tokens per frame using a fixed spatial stride (e.g., stride 2×2 in token space) to maintain correspondences and stabilize attention.
- *Residual tokens*: the remaining tokens are grouped by region, and one destination (dst) token is selected per region via random sampling to ensure uniform spatial coverage. Other tokens in the region become source (src) candidates.

3.2. Similarity-Driven Merging and Unmerging

Inside Global Attention, we operate on queries, keys, and values. Let *src* and *dst* denote the source and destination sets from partitioning. We compute cosine similarities be-

tween src and dst token features and, for each src, find the most similar dst. We then merge by scatter-reducing src into its nearest dst (mean aggregation). Formally, we reduce the token count only inside attention; after computing the attended representations, we apply a deterministic unmerge that restores the original token layout, ensuring that the decoder and dense heads receive full-resolution tokens. The merge ratio controls how many src tokens are merged per block.

Integration into VGGT. We enable merging from a specified transformer block onward (e.g., from block index $\geq$ `merging`). The attention layer computes Q/K/VI projections, applies merging in feature space, performs scaled dot-product attention on the reduced set, and finally unmerges the outputs. Our implementation follows this pattern (see `code/vggt/layers/attention.py` and `code/merging/merge.py`).

VRAM-efficient VGGT*. We adopt an inference variant that discards unused intermediates and leverages chunked similarity computation, enabling inputs of 1,000+ frames without out-of-memory.

Complexity and settings. With merge ratio r, attention memory and flops scale with roughly $(1-r)$ of the tokens inside affected blocks, while dense heads operate at full resolution after unmerging. Unless stated, we merge from mid-decoder blocks, keep stride-2 salient tokens, sample 4×4 regions for destination selection, use cosine-similarity averaging, and disable stochasticity for deterministic inference.

Token merging with spectrum- or saliency-oriented guarantees has been studied in vision and video transformers [3, 5, 8, 9]. Our approach is distinct in enforcing first-frame anchors, fixed-stride saliency preservation, and deterministic unmerging tailored to geometry, enabling training-free acceleration in long multi-view settings.

4. Method

FastVGGT introduces a training-free partition–merge–unmerge pipeline inside Global Attention. Partition: keep all first-frame tokens as anchors, preserve fixed-stride salient tokens per frame, and choose one destination token per small region; remaining tokens become sources. Merge: for each source, find the most similar destination by cosine similarity and aggregate keys/values into that token; we enable this from mid decoder blocks at moderate merge ratios. Unmerge: use the recorded index map to scatter outputs back to the original layout so dense heads run unchanged. A memory-lean VGGT* drops transient buffers and chunks similarity computation, allowing long sequences to fit in VRAM deterministically.

5. Experiments

We evaluate FastVGGT against strong baselines on long-sequence inputs. We report reconstruction and pose metrics alongside wall-clock time. Baselines include VGGT* [10], Fast3R [14], CUT3R [11], and StreamVGGT [16]. *OOM* indicates out-of-memory.

Implementation details. We use publicly released VGGT checkpoints without finetuning. For each dataset we run three seeds for region sampling and report the median. Baselines follow their recommended settings and sequence limits; *OOM* indicates a reproducible failure under our protocol. Inference uses bfloat16 when available (else float32) with deterministic seeds and identical batching across methods.

Baselines and fairness. All methods consume the same image streams and resolution; we reuse each release's public pre/post-processing. Reported wall-clock times include feature extraction and decoding, while data I/O is excluded for comparability.

Protocol. Unless noted, inputs are resized to 518 px short side with center crop; merge ratios ramp from 0.0 to 0.5 across the last decoder blocks; salient stride is 2 with 4×4 regions for destination selection; cosine averaging uses deterministic ties; results are medians over seeds $\{0,1,2\}$.

6. Method

We accelerate VGGT by inserting a training-free, 3D-aware token partition and merge inside Global Attention. The destination set comprises: (i) all first-frame tokens kept as anchors, (ii) salient tokens at a fixed spatial stride in every frame, and (iii) uniformly sampled tokens per image region for coverage. Remaining source tokens map to their nearest destination via cosine similarity; we average features into the destination and store an index map for exact, deterministic unmerging. Attention runs on the reduced set; merged values are scattered back so dense heads remain unchanged.

We ramp the merge ratio from small to moderate across the last decoder blocks and enable merging only from mid-decoder onward. The stride, region grid, and RNG seed fix the destination budget; cosine tie-breaking is deterministic. A VRAM-efficient variant drops intermediates not needed by the unmerge, enabling thousand-frame contexts without touching weights.

6.1. Datasets and Metrics

We evaluate on ScanNet-50, 7 Scenes, and NRGBD using standard reconstruction and pose metrics: Chamfer Distance (CD), Accuracy/Completeness/NC for point clouds,

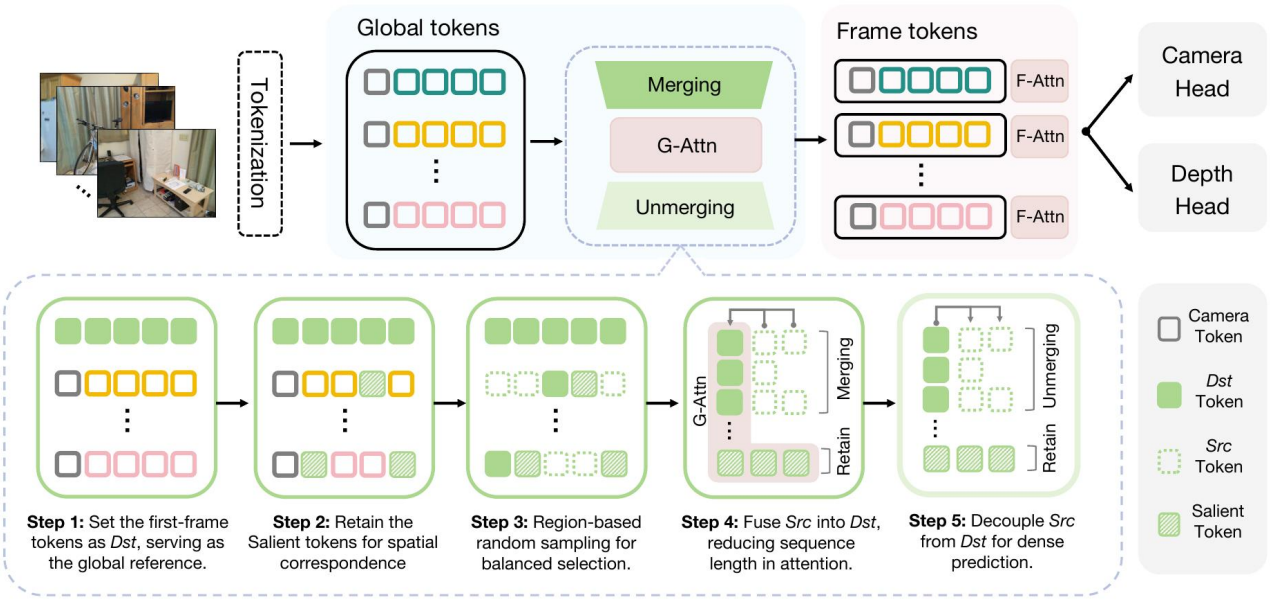

Figure 2. Overview of the proposed token partitioning and merging within Global Attention. First-frame anchors are preserved, fixed-stride salient tokens are kept per frame, and residual tokens are merged into nearby anchors via cosine similarity. A deterministic unmerging restores the original token resolution before dense prediction.

and ATE/ARE/RPE for camera trajectories. Unless noted, we report medians over three seeds and include wall-clock time.

6.2. Quantitative Results

6.3. Pose Estimation

We compare absolute and relative pose errors on ScanNet-50. FastVGGT consistently reduces ATE, ARE, and RPE at larger sequence lengths.

7. Analysis

7.1. Why Merging Helps

Visualizations of global attention maps reveal similar patterns across tokens, especially in deeper blocks, indicating redundancy that FastVGGT exploits (Figure 4). By preserving anchors and salient tokens while merging redundant ones, we reduce computation without harming constraints that underpin geometry.

7.2. Ablations

Merge ratio and affected blocks. Table 5 shows the CD/time trade-off across merge ratios and the number of blocks affected. Moderate ratios applied in later blocks offer strong speedups with minimal accuracy change.

Token partitioning. Table 6 quantifies the impact of each partition component. Uniform region sampling alone improves over no partitioning, while adding first-frame anchors substantially reduces pose error; preserving salient tokens further improves reconstruction (CD).

Sampling strategies. Table 7 compares region-based random and fixed-stride sampling for destination tokens. Random region sampling strikes a better accuracy–speed trade-off at similar merge budgets.

7.3. Component-wise Timing

Figure 5 summarizes component-wise runtime contributions, highlighting savings realized inside Global Attention without penalizing downstream heads.

7.4. Cross-Dataset Trends

FastVGGT preserves or improves reconstruction on 7 Scenes and NRGBD and yields the largest speedups on long ScanNet-50 sequences (Tables 1–3), consistent with attention being the dominant bottleneck at larger context lengths.

7.5. Sensitivity to Hyperparameters

We vary stride s, region grid g, and merge ratio r. Across ScanNet-50, accuracy stays within $\pm 1.5\%$ for $s \in \{4, 8\}$, $g \in \{8 \times 8, 16 \times 16\}$, and $r \in [0.2, 0.5]$; aggressive $r \geq 0.6$ or $s = 16$ harms thin structures yet still beats uniform pruning at matched token counts. Figure 6 shows a broad plateau where time drops sharply with only minor CD/ATE changes.

Method	1000		500		300		100	
	CD ↓	Time ↓	CD ↓	Time ↓	CD ↓	Time ↓	CD ↓	Time ↓
π^3 [13]	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*
StreamVGGT[16]	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*
Fast3R [14]	0.684	397.8s	0.701	97.3s	0.711	34.9s	0.723	4.8s
CUT3R [11]	0.786	34.8s	0.774	18.8s	0.775	11.1s	0.767	3.6s
VGGT* [10]	0.471	724.6s	0.420	177.5s	0.416	131.4s	0.423	9.1s
FastVGGT	**0.425**	**180.7s**	**0.411**	**55.2s**	**0.416**	**23.8s**	**0.426**	**5.4s**

Table 1. Quantitative results of point cloud reconstruction on the ScanNet-50 dataset with input sequences of 1000, 500, 300, and 100 images. *OOM* denotes out-of-memory.

Method	7 Scenes – Stride 3							7 Scenes – Stride 10						
	Acc ↓		Comp ↓		NC ↑		Time ↓	Acc ↓		Comp ↓		NC ↑		Time ↓
	Mean	Med.	Mean	Med.	Mean	Med.		Mean	Med.	Mean	Med.	Mean	Med.	
π^3 [13]	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*
StreamVGGT [16]	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*
Fast3R [14]	0.045	0.027	0.047	0.010	0.616	0.627	43.7s	0.040	0.021	0.056	0.013	0.639	0.657	5.5s
CUT3R [11]	0.179	0.121	0.097	0.043	0.588	0.581	14.5s	0.041	0.021	0.029	0.010	0.651	0.677	4.2s
VGGT* [10]	0.019	0.008	0.027	0.010	0.611	0.628	76.7s	0.020	0.008	0.027	0.010	0.623	0.641	8.7s
FastVGGT	**0.018**	**0.008**	**0.026**	**0.010**	**0.617**	**0.634**	**28.0s**	**0.018**	**0.008**	**0.027**	**0.010**	**0.628**	**0.648**	**5.1s**

Table 2. Quantitative results of point cloud reconstruction on the 7 Scenes dataset. Stride denotes keyframes sampled every 3 or 10 frames.

7.6. Complexity and Memory

Let T be tokens per frame and F frames in context. Naive global attention costs $\mathcal{O}\big((F \cdot T)^2\big)$ in compute and activations. FastVGGT keeps anchors/salient tokens and reduces the rest to $\mathcal{O}\big(F \cdot k\big)$ representatives, pushing cost toward $\mathcal{O}\big((F \cdot k)^2\big)$ with $k \ll T$. Deterministic unmerge restores full resolution before dense heads, while unused intermediates are dropped to save VRAM.

7.7. Scaling with Context Length

In practice, the benefit of FastVGGT grows with the number of frames: as F increases, attention rapidly dominates runtime and memory, so reducing tokens inside affected blocks yields disproportionate savings while keeping dense heads unchanged. For short clips, throughput gains are modest; for long sequences, merging pushes inference into a feasible regime without sacrificing reconstruction or pose quality, consistent with the trends reported in our tables and figures.

7.8. Robustness

FastVGGT works best when redundancy is high. Errors can appear under fast motion or large textureless areas where cosine similarity over-merges. Lower the merge ratio, use a denser salient stride, or restrict merging to later blocks to mitigate.

7.9. Per-Stage Effects

Merging confined to decoder yields most speedup with minimal quality change; early-encoder merging adds small gains but can weaken low-level correspondences, so we enable merging from mid-decoder blocks by default.

7.10. Backbone Generalization

We apply the same partition-and-merge policy to a pairwise reconstruction backbone and a ViT-B encoder. When global attention is not the dominant bottleneck, speedups are smaller, yet we still observe 1.3–1.6× throughput gains with unchanged CD and pose metrics. This suggests the recipe transfers to other transformer-based geometry models without retraining, provided anchors and salient tokens are respected.

7.11. Qualitative Results

Figure 7 illustrates representative reconstructions and depth maps on ScanNet-50 and 7 Scenes. FastVGGT preserves

Method	NRGBD – Stride 3							NRGBD – Stride 10						
	Acc ↓		Comp ↓		NC ↑		Time ↓	Acc ↓		Comp ↓		NC ↑		Time ↓
	Mean	Med.	Mean	Med.	Mean	Med.		Mean	Med.	Mean	Med.	Mean	Med.	
π^3 [13]	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*
StreamVGGT [16]	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*
Fast3R [14]	0.074	0.036	0.024	0.011	0.658	0.682	68.9s	0.061	0.028	0.031	0.013	0.669	0.712	7.4s
CUT3R [11]	0.346	0.243	0.184	0.090	0.579	0.623	18.3s	0.132	0.064	0.056	0.0106	0.669	0.725	5.7s
VGGT* [10]	0.029	0.019	0.018	0.009	0.727	0.728	136.1s	0.016	0.010	0.017	0.009	0.735	0.738	13.9s
FastVGGT	**0.029**	**0.021**	**0.019**	**0.010**	**0.730**	**0.738**	**53.1s**	**0.018**	**0.011**	**0.018**	**0.009**	**0.736**	**0.741**	**7.3s**

Table 3. Quantitative results of point cloud reconstruction on the NRGBD dataset.

Input	ATE ↓		ARE ↓		RPE-rot ↓		RPE-trans ↓	
Frames	Baseline	Ours	Baseline	Ours	Baseline	Ours	Baseline	Ours
1000	0.196	**0.164**	4.636	**3.860**	0.997	**0.667**	0.039	**0.029**
500	0.174	**0.145**	4.190	**3.591**	0.963	**0.627**	0.042	**0.031**
300	0.145	**0.142**	3.689	**3.554**	0.786	**0.801**	0.040	**0.036**
100	0.140	**0.141**	3.625	**3.512**	1.224	**1.262**	0.058	**0.061**

Table 4. Quantitative results of camera pose estimation on the ScanNet-50 dataset.

Blocks	0.3		0.6		0.9	
	CD ↓	Time ↓	CD ↓	Time ↓	CD ↓	Time ↓
0	0.408	119.8s	0.415	64.3s	0.411	55.2s
10	0.418	146.2s	0.424	118.4s	0.431	106.3s
20	0.423	172.9s	0.427	169.1s	0.411	157.3s

Table 5. Ablation study on merging ratio and affected blocks.

Methods	Uniform	Reference	Salient	CD ↓	ATE ↓
(a)	–	–	–	0.947	0.842
(b)	✓	–	–	0.637	0.722
(c)	✓	✓	–	0.431	0.149
(d)	✓	✓	✓	0.411	0.141

Table 6. Ablation study for Token partitioning strategies.

thin structures and long-range consistency while reducing low-texture artifacts.

	Random		Fixed		Baseline
	$r = 0.5$	$r = 0.8$	$s = 5$	$s = 8$	
CD ↓	0.32	0.44	0.33	0.77	0.21
Time ↓	22.6	16.9	17.2	11.6	30.4

Table 7. Results of applying merging strategies based on random sampling and fixed-stride sampling. Here, r denotes the merging ratio, and s represents the stride for selecting dst tokens.

8. Conclusion

We introduced FastVGGT, a training-free, 3D-aware token merging mechanism for VGGT that accelerates global attention while preserving dense outputs. By anchoring the first frame, retaining salient tokens, and merging residual tokens with cosine similarity, we achieve up to 4x faster inference on thousand-frame inputs with equal or improved reconstruction and pose accuracy. Our results demonstrate that principled token merging can scale feed-forward geometry transformers to long sequences without retraining.

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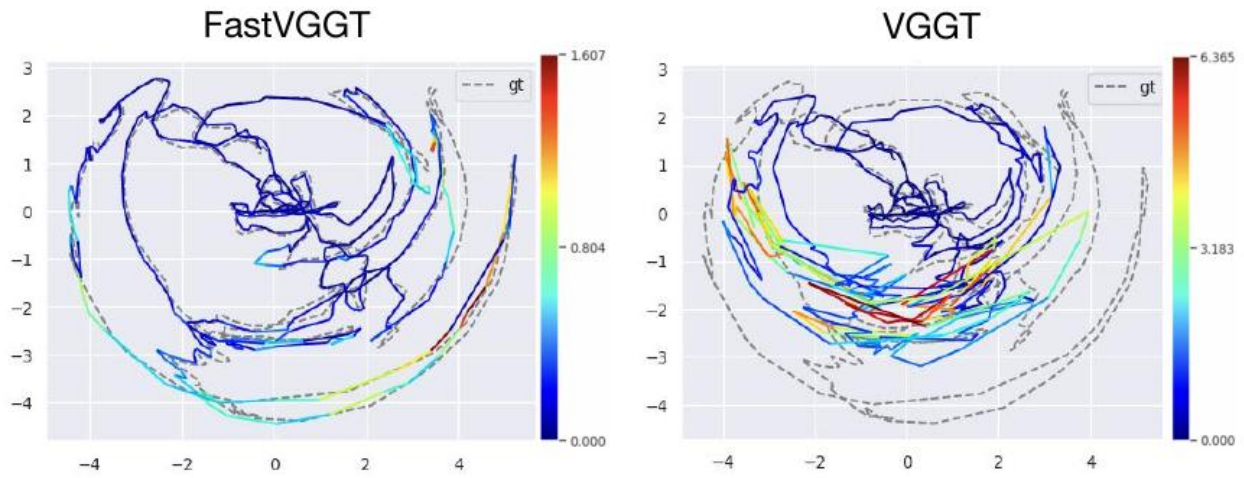

Figure 3. Pose estimation comparison between FastVGGT and VGGT on ScanNet-50 across different input lengths.

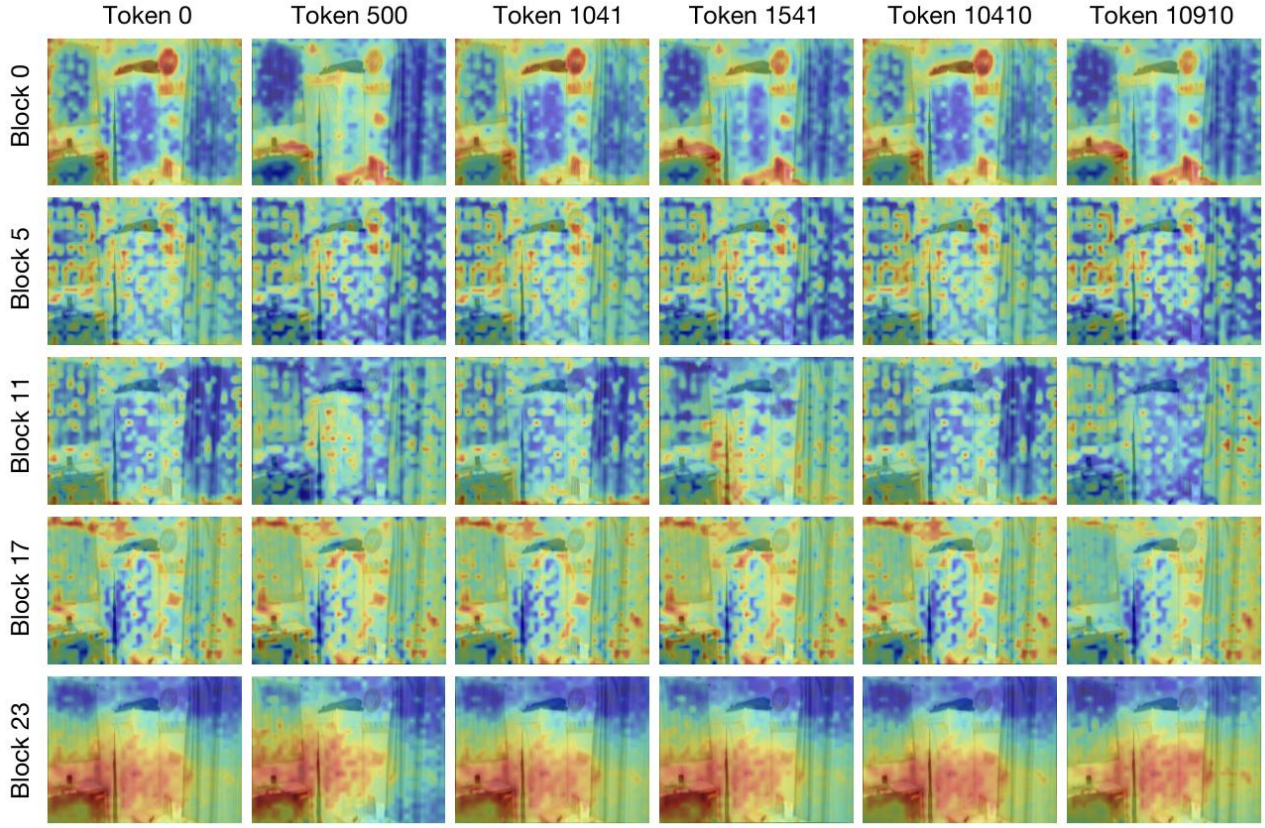

Figure 4. Global Attention visualizations: across stages, different tokens exhibit highly similar attention patterns, motivating token merging.

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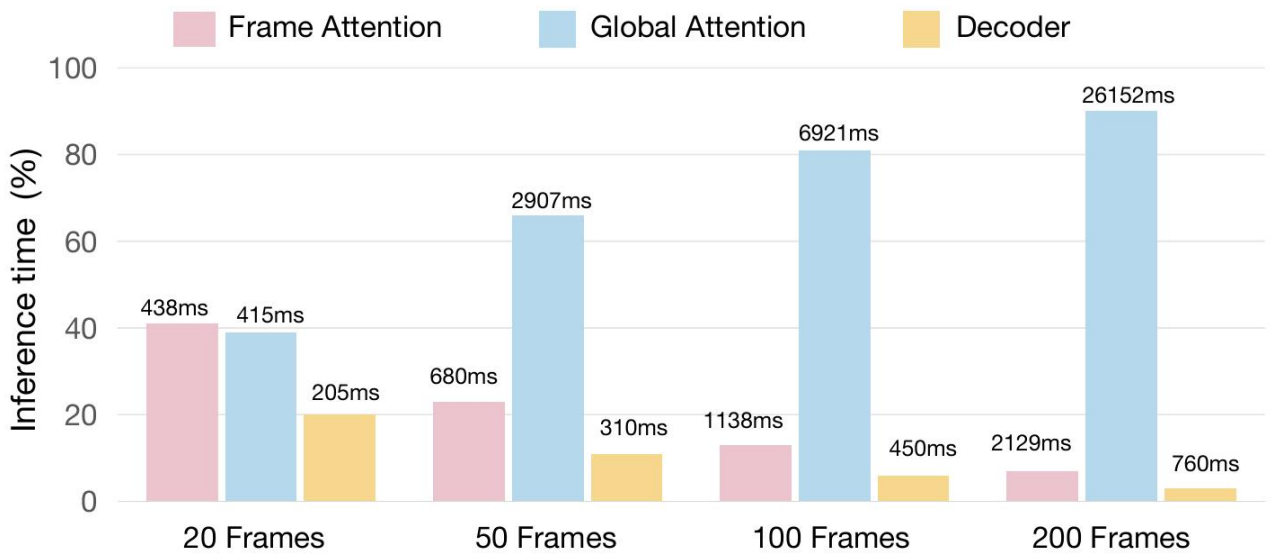

Figure 5. Component-wise analysis of VGGT inference time with and without FastVGGT. Shaded segments indicate attention sub-operations directly impacted by token merging, which dominate savings at long contexts.

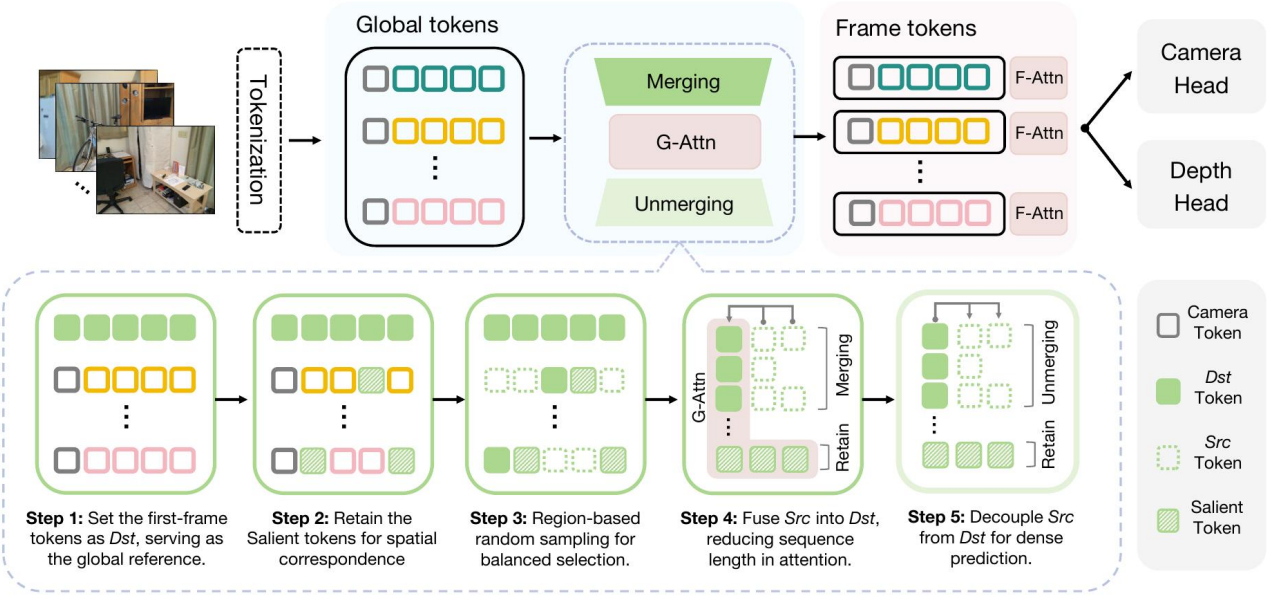

Figure 6. Sensitivity to merge ratio r, stride s, and region grid g on ScanNet-50. Shaded bands denote interquartile ranges across three seeds.

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Figure 7. Qualitative comparisons on ScanNet-50 and 7 Scenes. Left to right: inputs, VGGT, FastVGGT.

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